PHARMACOVIGILANCE

Virtual Task – Summary of Adverse Drug Reactions & Pharmacovigilance

Reference video: Understanding Adverse Drug Reactions & Pharmacovigilance – Dr. Nirmal George

1. Introduction

Pharmacovigilance is the science and activities concerned with the detection, assessment, understanding and prevention of adverse effects or any other medicine-related problem. Its main purpose is to improve patient safety and ensure that the benefits of medicines continue to outweigh their risks.

2. Adverse Drug Reaction (ADR)

An adverse drug reaction is a harmful and unintended response to a medicinal product when the medicine is used at normal doses for prevention, diagnosis or treatment. ADRs are important because they can increase illness, hospital admissions, treatment costs and, in serious cases, may be life-threatening.

3. Important Features of Pharmacovigilance

Activity	Purpose
Detection	Identify suspected adverse effects and safety signals.
Assessment	Evaluate the seriousness, frequency and possible relationship with a medicine.
Understanding	Study patterns, risk factors and mechanisms of medicine-related harm.
Prevention	Take appropriate measures to reduce avoidable medicine-related risks.
Communication	Share relevant safety information with healthcare professionals and patients.

4. Classification of ADRs

Type A (Augmented): Dose-related and generally predictable reactions caused by the known pharmacological action of a medicine.
Type B (Bizarre): Unpredictable reactions that are usually not dose-related, such as certain allergic or idiosyncratic reactions.
Other categories: ADRs may also be described as chronic, delayed, end-of-use reactions, or failure of therapy depending on their timing and characteristics.

5. Why ADR Reporting is Important

Clinical trials are performed in relatively limited populations and may not detect every rare or delayed adverse effect. Post-marketing surveillance therefore provides important real-world information. Reporting suspected ADRs helps identify previously unrecognized safety signals and supports appropriate regulatory and clinical action.

6. Role of Healthcare Professionals

Doctors, pharmacists, nurses and other healthcare professionals contribute to pharmacovigilance by recognizing suspected ADRs, documenting relevant patient and medicine information, reporting suspected reactions through appropriate reporting systems, and educating patients about safe medicine use. Pharmacists have an important role because they routinely monitor medicines, interactions, adherence and possible adverse effects.

7. Key Takeaways

• Pharmacovigilance focuses on medicine safety throughout the medicine's life cycle.

• ADRs are harmful and unintended responses associated with normal use of medicines.

• Early recognition and reporting can help detect important safety signals.

• Post-marketing surveillance complements evidence obtained from clinical trials.

• Effective communication and reporting by healthcare professionals improve patient safety.

Conclusion: Pharmacovigilance is an essential part of safe and rational medicine use. Systematic monitoring and reporting of adverse drug reactions help identify risks, prevent avoidable harm and maintain a favourable benefit–risk balance for patients.

Prepared as a virtual-task summary based on the referenced video and standard pharmacovigilance concepts.